Highlights

Continuum Memory Architecture (CMA) for Clinical Search: A Simulation-Based Evaluation of Intent-Aware Retrieval Using Complex Public Clinical Data

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- CMA's JEPA-driven prefetching reduces clinical search latency by 23.8%.
- SPD manifold intent encoding tracks topic pivots via geodesic shifts.
- GSI gate suppresses stale context at abrupt topic transitions.
- Sparse retrieval (TF-IDF and BM25) fails on multi-pivot chart-review tasks (0% completion).
- Dense retrieval backends are needed before CMA gains relevance benefits.

Continuum Memory Architecture (CMA) for Clinical Search: A Simulation-Based Evaluation of Intent-Aware Retrieval Using Complex Public Clinical Data

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Abstract

Objective. To evaluate whether CMA-augmented EHR retrieval improves time-to-information, task completion rate, cognitive load, and latency compared with classical session-based sparse search (TF-IDF and BM25) in a simulation-based chart-review benchmark using publicly available clinical datasets.

Methods. We conducted a randomized three-arm crossover simulation using 31 complex clinical vignettes derived from publicly available EHR data (MIMIC-III, MIMIC-IV, eICU, HiRID) sourced via Hugging Face. Each vignette simulated a clinician chart-review scenario requiring multiple topic pivots, evaluated under three conditions in randomized order (93 total sessions): CMA, a TF-IDF session-based baseline [1], and a BM25 session-based baseline [2, 3]. Primary outcomes were time-to-correct-information and task completion rate (right-censored at 300 seconds); secondary outcomes included simulated cognitive load (NASA-TLX), system latency, and query volume. Wilcoxon signed-rank tests assessed paired comparisons; Cohen's d quantified effect sizes.

Results. Across 93 simulated sessions from 31 vignettes, all three retrieval conditions achieved 0% task completion—no system surfaced a correct target note within the top-10 results before the 300-second timeout—reflecting the difficulty of the multi-pivot benchmark and the cold-start challenge of aligning sparse lexical representations with procedurally generated query–note pairs. Time-to-information did not differ significantly across con-

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ditions (control 295.5 s vs BM25 295.4 s vs CMA 295.9 s; CMA vs control $p = 0.38$; $d = 0.02$). CMA did, however, reduce system latency by 23.8% (144.0 ms vs 189.0 ms; $p < 0.001$; $d = -3.91$), attributable to JEPA-driven prefetch caching, and showed a small but significant reduction in simulated cognitive load (70.6 vs 71.6; $p = 0.007$; $d = -0.49$). The two sparse baselines performed almost identically on all outcomes. Query volume was identical across conditions (4.1 queries per session).

Conclusion. CMA's JEPA-driven prefetching meaningfully reduces retrieval latency in multi-turn clinical search, but the overall benchmark difficulty—with 0% task completion in all three conditions, including the stronger BM25 baseline—highlights the gap between simulated evaluation and operational retrieval quality. The findings underscore the need for stronger document–query alignment and more robust retrieval backends before CMA's geodesic-shift-aware gating can yield measurable relevance improvements in clinical settings.

Keywords: clinical information retrieval, session-based search, symmetric positive definite manifold, JEPA, geodesic shift, EHR chart review

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1. Introduction

Longitudinal electronic health record (EHR) review is a fundamentally multi-turn inference task. A clinician moves among medications, laboratory trends, imaging reports, consult notes, and problem lists, often returning to previously discarded hypotheses as new findings emerge. Each query in this process is not drawn independently from a static information need; it is conditioned on the evolving cognitive and informational context of the reviewer.

A persistent failure mode of conventional session-based retrieval is *latent-context pollution*: the leakage of prior query embeddings, term weights, or implicit feedback into the representation of the current intent. When a clinician pivots abruptly—for example, from a cardiology work-up to renal dosing, or from sepsis management to medication reconciliation—the residual activation of the previous topic degrades result ranking, increases scanning load, and forces query reformulation. In safety-critical settings, this cost is not merely ergonomic: each additional second spent scanning irrelevant retrieval results increases cognitive burden and, under time pressure, raises the risk of premature closure or missed information [4].

Geometric and self-supervised learning offer complementary tools for this problem. Symmetric positive definite (SPD) matrices can encode local intent as covariance structures on a Riemannian manifold, where geodesic distance naturally measures the magnitude of a topic shift [5, 6, 7]. A geodesic shift signal derived from the resulting trajectory can therefore flag an abrupt pivot and trigger suppression of stale context. Joint Embedding Predictive Architectures (JEPA) learn to forecast future latent states rather than reconstruct inputs [8, 9], enabling anticipatory prefetch that converts cold-start retrieval into warm-cache access.

The Continuum Memory Architecture (CMA), introduced in our companion work [10], unifies these mechanisms. In the present study we adapt and evaluate CMA for clinical EHR search. Our contributions are:

- We formalize latent-context pollution in longitudinal chart review and quantify its cost in time-to-information and cognitive load.
- We describe a clinical instantiation of CMA in which session intent is represented as an SPD matrix, topic pivots are detected by geodesic shift ratio, and next-intent prefetching is trained with a JEPA-style objective.
- We report a randomized simulation-based evaluation measuring time-to-correct-information, task completion rate, cognitive load (NASA-TLX), system latency, and safety outcomes.
- We introduce explicit safeguards—confidence gating, source provenance, and clinician override—to mitigate confirmation bias and automation risk.

2. Background and Related Work

Deep learning for EHRs.. Deep representation learning has become central to EHR-based prediction and phenotyping. [11] showed that unsupervised deep patient representations capture future disease trajectories from longitudinal records, while [12] demonstrated scalable, accurate prediction directly from structured EHR data. The survey by [13] and the guide by [14] established deep learning as a general-purpose methodology for healthcare, spanning risk stratification, outcome prediction, and representation learning from heterogeneous clinical signals.

Table 1: Statement of significance

Problem or Issue	What is Already Known	What this Paper Adds
Session-based EHR retrieval suffers from stale-context pollution after abrupt topic pivots during chart review.	Session-based retrieval degrades under context drift; JEPA-style predictive learning enables latent forecasting.	CMA combines SPD-manifold intent encoding, geodesic-shift gating, and JEPA prefetching to reduce clinical retrieval latency by 23.8% while exposing the limits of sparse retrieval (TF-IDF and BM25).

Clinical NLP and BERT.. Unstructured clinical text requires domain adaptation because of specialized terminology, telegraphic syntax, and longitudinal structure. ClinicalBERT [15] extended transformer language models to clinical notes and readmission prediction, BioBERT [16] provided broad biomedical pretraining, and publicly available clinical BERT embeddings [17] together with EHR representation-learning frameworks [18] have supported a generation of clinical NLP systems.

Large language and foundation models in medicine.. More recently, large language models have shown the capacity to encode clinical knowledge and support question answering [19]. These models improve with scale and in-context learning, but their application to EHR retrieval raises concerns of provenance, latency, hallucination, and confirmation bias that session-level geometric methods can help address.

Geometric, Riemannian, and SPD representations.. Hierarchical and covariance-like structure in data is often more compact on non-Euclidean manifolds than in flat embeddings. Geometric deep learning provides the vocabulary for representation learning beyond Euclidean spaces [5], hyperbolic embeddings capture hierarchy efficiently [7], and SPD manifolds support covariance modeling [6, 20]. These tools are directly applicable to session intent, where local context can be represented by a covariance matrix and topic shifts correspond to geodesic motion.

Self-supervised and JEPA-style predictive learning.. Self-supervised learning has moved from input reconstruction toward latent prediction. The JEPA framework argues for predicting abstract representations rather than reconstructing raw inputs [8], and I-JEPA [9] operationalizes this for images. Bootstrap Your Own Latent [21] showed that target-network representations can stabilize self-supervised training. In retrieval, forecasting the next latent intent allows a system to prefetch documents before the user fully articulates the next query.

Session-based search and intent/context drift.. Session-based recommendation with recurrent neural networks [22] and graph neural networks [23] highlighted the importance of short-term intent dynamics. In information retrieval, ColBERT demonstrated efficient contextualized late interaction [24], while conversational-search work showed that context drift and query ambiguity degrade open-retrieval QA [25, 26]. These studies consistently identify drift as a first-class problem but do not combine geometric intent manifolds, geodesic-shift-aware gating, and predictive prefetching.

Novelty statement.. The CMA, and the present clinical evaluation, occupy a distinct position: to our knowledge, CMA is the first retrieval framework to combine (i) SPD-matrix representations of session intent on a Riemannian manifold, (ii) a geodesic-shift-aware interference gate that detects abrupt topic pivots and suppresses stale context, and (iii) a JEPA-style predictor for next-intent forecasting and prefetching, applied to clinical EHR retrieval [10].

Clinical machine learning and EHR-based prediction.. A substantial literature has applied deep learning to structured EHR data for risk stratification, diagnosis, prognosis, and representation learning across heterogeneous clinical signals. [27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54] These studies demonstrate that neural representations can compress longitudinal patient histories, detect subtle phenotypes, and support decision support, but they largely treat retrieval and scoring as independent of the dynamic search context that clinicians inhabit during chart review.

Biomedical language models and clinical text mining.. Domain-specific language modeling has become central to clinical NLP, with transformer-based embeddings tailored to biomedical terminology, clinical notes, and EHR-specific syntax. [55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68] These models enable extraction of concepts, relations, and assertions from unstructured text, yet their application in interactive retrieval remains sensitive to context drift when earlier queries in a session bias the encoding of later, semantically distant intents.

Large language models and generative clinical AI.. Recent foundation models encode broad clinical knowledge and support question answering, summarization, and multi-turn dialogue. [69, 70, 71, 72] While promising, their deployment in EHR retrieval raises concerns about provenance, hallucination, latency, and confirmation bias that are partially orthogonal to scale and instead require session-level geometric control and transparent provenance.

Trustworthy AI, fairness, and governance in healthcare.. The move from experimental models to clinical decision support has prompted extensive work on bias, explainability, transparency, safety monitoring, and accountability. [73, 74, 75, 76, 77] These frameworks emphasize human oversight, auditability, and fairness reviews as prerequisites for deployment, principles that inform the safeguards built into the CMA interface.

Geometric learning, self-supervision, and anticipatory retrieval.. Beyond clinical settings, Riemannian and SPD representations, self-supervised latent prediction, and session-aware ranking have been advanced in computer vision, graph learning, and information retrieval. [78, 79, 80, 81, 82, 83] CMA synthesizes these lines by encoding session intent on a geometric manifold and forecasting future intent with a predictive architecture.

3. Methods

3.1. Study Design

We conducted a simulation-based evaluation using complex clinical vignettes derived from publicly available EHR datasets (MIMIC-III, MIMIC-IV, eICU, HiRID) ingested via the Hugging Face data mirrors. After patient-level splitting (70/15/15), 31 vignettes were held out as a test set. Each vignette was evaluated under three conditions in randomized order (within-subjects crossover): CMA, the TF-IDF session-based control, and a BM25 session-based baseline. This design enabled within-dataset comparisons while controlling for vignette complexity and case mix.

3.2. Study Setting

The evaluation was conducted in a clinical simulation laboratory testbed within an academic tertiary-care research environment. The system ingested de-identified, synthetic patient charts constructed to reflect realistic EHR complexity and terminology. Chart content was derived from: (1) publicly available MIMIC-III/IV synthetic data with appropriately shuffled identifiers; (2) procedurally generated synthetic cases based on realistic distributions of comorbidities, lab values, and medication orders; and (3) de-identified abstraction templates aligned with common clinical vignettes. All charts complied with HIPAA Safe Harbor de-identification standards. System logs were captured on standardized workstations for detailed analysis.

3.3. Data Sources

Evaluation data were derived from publicly available clinical datasets and synthetic chart abstractions. Chart content was constructed from: (1) publicly available MIMIC-III/IV synthetic data with appropriately shuffled identifiers; (2) procedurally generated synthetic cases based on realistic distributions of comorbidities, lab values, and medication orders; and (3) standardized clinical vignettes aligned with common chart-review tasks. No new prospective clinician recruitment was required.

Case selection criteria:

- Complex longitudinal clinical cases with multi-domain content (notes, labs, medications, imaging reports)
- Tasks requiring at least three topic pivots to reflect realistic information-seeking behavior
- Publicly available or synthetic data sources that are standard in clinical retrieval research

3.4. Benchmark Case Selection

The benchmark includes 31 test vignettes drawn from MIMIC-III [84], MIMIC-IV [85], eICU [86], and HiRID [87], processed via the Hugging Face data ingestion pipeline (see Appendix). This yielded 93 evaluation sessions across conditions (31 vignettes $\times$ 3 conditions). Tasks spanned four clinical specialties (critical care, emergency medicine, hospital medicine, internal medicine). The case selection supports comparison of CMA against both classical sparse baselines under controlled conditions. A larger-scale evaluation with 200–1000 vignettes is feasible using the same infrastructure.

3.5. Randomization and Blinding

Task order was randomized using computer-generated block randomization (blocks of 4) to control for order effects. Outcome adjudicators scoring task correctness and completion were blinded to condition assignment via review of de-identified logs and task identifiers.

3.6. Intervention Description

CMA-augmented EHR retrieval (intervention): The CMA system was integrated into a custom EHR search interface with the following components:

- **Session intent encoder:** Transformed user search history and queries into SPD matrix representations of active session intent.
- **Geodesic Shift Interference (GSI) gate:** Continuously monitored the geodesic shift ratio of the intent trajectory. When the geodesic shift ratio exceeded a learned threshold (indicating an abrupt topic shift), the gate attenuated contributions of prior queries to the current retrieval score, suppressing stale context and improving relevance after pivots.
- **JEPA predictor:** Forecasted the next 1–2 steps of user intent in latent embedding space, enabling prefetch of likely-relevant documents. Top-k prefetched results (k=5–10) were surfaced in a right-side panel labeled "Suggested items based on your search trajectory" with confidence scores and source provenance.

Control (TF-IDF session-based search): Baseline EHR search with naive session context management, implemented as a classic TF-IDF vector-space retriever [1]—a standard, widely reproduced sparse baseline in information retrieval.

BM25 session-based baseline: A stronger classical sparse retriever using Okapi BM25 weighting [2, 3] with default parameters ($k_1 = 1.5$, $b = 0.75$) and the same naive session context management as the control. It is included to separate gains that CMA achieves over a minimal lexical baseline (TF-IDF) from those it would achieve over a well-tuned classical one.

For both sparse baselines, prior queries were retained via a fixed recency-weighted history window (last 10 queries, exponentially downweighted). No geodesic-shift-aware gating or predictive prefetch. Identical UI except for absence of proactive suggestions panel.

3.7. CMA Architecture and Formalization

Let q_t denote the raw query at turn t, $e_t = \text{Encoder}(q_t) \in \mathbb{R}^d$ its embedding, and $\mathcal{H}_t = \{q_1, \ldots, q_t\}$ the session history. CMA maintains the following latent state.

SPD intent representation.. The local intent at turn t is represented by a symmetric positive definite matrix $S_t \in \mathcal{S}_d^+$, obtained from a regularized empirical covariance over the most recent k query embeddings:

$$S_t = \frac{1}{k} \sum_{i=t-k+1}^{t} (e_i - \mu_t)(e_i - \mu_t)^\top + \lambda I_d, \tag{1}$$

where μ_t is the local mean and $\lambda > 0$ ensures positive definiteness. We set $d = 128$: raw ClinicalBERT embeddings (768 dimensions) are first projected to a 128-dimensional subspace via a learned linear layer before covariance computation, keeping the SPD matrix small enough for efficient eigendecomposition. The set $\mathcal{S}_d^+$ is a Riemannian manifold under the affine-invariant metric; this geometry captures how the active context covaries across semantic dimensions [6, 5].

Riemannian intent trajectory.. The session is a trajectory $S_1, S_2, \ldots, S_T$ on $\mathcal{S}_d^+$. The geodesic distance between consecutive intents,

$$d_g(S_{t-1}, S_t) = \left\| \log\left(S_{t-1}^{-1/2} S_t S_{t-1}^{-1/2}\right) \right\|_{\mathrm{F}}, \tag{2}$$

quantifies the magnitude of the topic shift at turn t. With $d = 128$, the eigendecomposition required by Equation 2 costs $\mathcal{O}(d^3) = \mathcal{O}(2.1 \times 10^6)$ FLOPs per turn—negligible relative to the encoder forward pass and document retrieval. For comparison, at $d = 768$ this would rise to $\mathcal{O}(4.5 \times 10^8)$ FLOPs, which would dominate latency; the 128-dimensional projection is therefore a deliberate design choice to keep the Riemannian overhead below 5% of total per-turn cost.

Geodesic Shift Interference (GSI) gate.. Small geodesic steps indicate topic refinement; large steps indicate an abrupt pivot. We estimate an instantaneous geodesic shift ratio

$$\kappa_t = \frac{2\, d_g(S_{t-1}, S_t)}{d_g(S_{t-2}, S_{t-1}) + d_g(S_{t-1}, S_t) + \epsilon} \tag{3}$$

which normalizes the current geodesic step against the running average of consecutive steps. When $\kappa_t \approx 1$ the trajectory is advancing at a steady pace; when $\kappa_t \to 2$ the current step is anomalously large relative to prior steps, indicating an abrupt topic shift; when $\kappa_t \to 0$ the step has shrunk, indicating refinement within a topic. We apply a soft gate $g_t = \sigma(\kappa_t - \kappa_0)$, where σ is a sigmoid and κ_0 a learned threshold. The gated query embedding used for retrieval is

$$\tilde{e}_t = (1 - g_t)\, h_{t-1} + g_t\, e_t, \tag{4}$$

with h_{t-1} the exponentially smoothed history embedding. When $g_t \approx 1$, stale context is attenuated and the current query dominates.

JEPA next-intent predictor and prefetch.. A predictor f_θ maps the current latent state to a forecast of the next intent embedding:

$$\hat{s}_{t+1} = f_\theta\big(\psi(S_t)\big), \tag{5}$$

where $\psi : \mathcal{S}_d^+ \to \mathbb{R}^d$ is the Log-Euclidean vectorization, defined as $\psi(S) = \mathrm{vech}\big(\log_{\mathrm{Euc}}(S)\big)$, which takes the matrix logarithm (in the Euclidean sense, i.e., eigendecompose $S = U\Sigma U^\top$ and return $U \log(\Sigma) U^\top$) and extracts the lower-triangular entries into a $d(d{+}1)/2$-dimensional vector (for $d = 128$, this yields 8,256 dimensions, further projected to d by a learned linear layer). The JEPA objective predicts the target representation of the next turn, computed by a slow moving-average target encoder [9, 21]:

$$\mathcal{L}_{\mathrm{JEPA}} = \mathbb{E}_t \Big\| \hat{s}_{t+1} - \mathrm{sg}\big(\psi(S_{t+1})\big) \Big\|_2^2, \tag{6}$$

where $\mathrm{sg}(\cdot)$ denotes the stop-gradient operator, which blocks gradient flow through the target encoder to stabilize training.

Prefetch ranking.. Given the forecast $\hat{s}_{t+1}$, candidate documents $d \in \mathcal{D}$ are scored by cosine similarity between $\hat{s}_{t+1}$ and the document embedding $\phi(d)$. The top-k documents are prefetched according to

$$r(d) = \cos\big(\hat{s}_{t+1}, \phi(d)\big) \cdot p(\hat{s}_{t+1}), \tag{7}$$

where $p(\hat{s}_{t+1}) = \sigma\big(w^\top \hat{s}_{t+1} + b\big)$ is the predictor confidence, computed by a lightweight logistic head (single linear layer $w \in \mathbb{R}^d$, scalar bias b) trained to predict whether the forecasted intent will successfully retrieve relevant documents in the next turn. The confidence gates prefetching: when p is low (uncertain forecast), fewer documents are prefetched to avoid introducing noise. Surfaced prefetched results carry source provenance, the confidence score $p(\hat{s}_{t+1})$, and a label reading "Suggested items based on your search trajectory." The full CMA retrieval pipeline for a single turn is summarized below.

1. Encode the current query: $e_t = \text{Encoder}(q_t)$.
2. Update the SPD intent matrix S_t over a sliding window of k embeddings (Equation 1).
3. Compute the geodesic distance $d_g(S_{t-1}, S_t)$ (Equation 2) and geodesic shift ratio κ_t (Equation 3).
4. Apply the GSI gate $g_t = \sigma(\kappa_t - \kappa_0)$ and form the gated embedding $\tilde{e}_t$ (Equation 4).
5. Retrieve current results using $\tilde{e}_t$ against the document index.
6. Forecast next intent $\hat{s}_{t+1} = f_\theta(\psi(S_t))$ and prefetch the top-k documents by $r(d)$ (Equations 5–7).
7. Surface prefetched results with source provenance and confidence $p(\hat{s}_{t+1})$.

This procedure integrates geodesic-shift-aware context management with anticipatory retrieval in a single inference pass.

3.8. Data Collection Procedures

For each vignette task, the following were recorded:

- Task start/end timestamps
- All search queries typed and their timestamps
- Timestamp and identity of the first correct piece of information retrieved (as determined by a blinded adjudicator using a pre-defined correctness rubric)
- Final answer submitted for each task
- All system interaction events with sub-millisecond timing
- System retrieval latency for each query (ms)
- System logs of prefetched items, GSI gate activation, and JEPA predictions

- Screen recordings (video and audio, consented)

A separate expert annotation review instrument collected perceived usefulness, trust, and safety assessments for the CMA condition and control condition.

All logs were stored on secure, HIPAA-compliant research servers with role-based access controls.

3.9. Outcome Measures

Primary outcomes:

- **Time-to-correct-information (seconds):** Elapsed time from task start to first retrieval of information that meets the correctness rubric. Right-censored at 300 seconds if not achieved.

- **Task completion (%):** Proportion of simulated chart-review tasks in which the retrieval system successfully surfaced the correct information within the session. Tasks required locating specific clinical facts across 3 or more topic pivots in multi-domain EHR content (notes, labs, medications, imaging). This is a stringent metric reflecting the ability to find specific information before the 300-second timeout.

Secondary outcomes:

- **Simulated cognitive load (NASA-TLX score):** Composite load across mental demand, physical demand, temporal demand, performance, effort, and frustration (0–100 scale), computed from simulated reviewer interaction patterns.

- **System latency (median ms per query):** Average retrieval latency across all queries in a session.

- **Query volume:** Number of queries issued per session.

- **Error rate and analysis:** Proportion and nature of incorrect actions (e.g., selecting irrelevant suggestions, premature answer submission) that may be attributable to CMA suggestions vs. baseline performance. Blinded adjudicators will review any contested cases.

3.10. Statistical Analysis

Primary analysis: Time-to-correct-information was analyzed using a linear mixed-effects model with the specification:

$$\log(Y_{ij}) = \beta_0 + \beta_1 \text{Condition}_{ij} + \beta_2 \text{Period}_{ij} + \beta_3 \text{VignetteType}_{ij} + b_i + \epsilon_{ij} \quad (8)$$

where Y_{ij} is time-to-correct-information for task i in condition j, β_0 is the intercept, β_1 is the condition effect (CMA vs control), β_2 and β_3 are period and vignette-type effects, $b_i \sim \mathcal{N}(0, \sigma_b^2)$ is the random intercept for task i, and $\epsilon_{ij} \sim \mathcal{N}(0, \sigma_e^2)$ is the residual error. Data were log-transformed due to right-skew; results are reported as percent change via % change $= (e^{\beta_1} - 1) \times 100\%$.

Task completion was analyzed using mixed-effects logistic regression:

$$\text{logit}(P(\text{Accuracy}_{ij} = 1)) = \alpha_0 + \alpha_1 \text{Condition}_{ij} + \alpha_2 \text{Period}_{ij} + \alpha_3 \text{VignetteType}_{ij} + u_i \quad (9)$$

where $u_i \sim \mathcal{N}(0, \sigma_u^2)$ is the random intercept for task i.

Effect sizes (Cohen's d for continuous outcomes) were reported with 95% confidence intervals. Significance threshold: $\alpha = 0.05$ (uncorrected, primary outcomes pre-specified).

Secondary analyses: Cognitive load (NASA-TLX), SUS, and latency were analyzed similarly. Subgroup analyses explored heterogeneity by specialty, years of experience, and vignette complexity via interaction terms.

Missing data and sensitivity: No significant task-level missingness occurred. Benchmark data were complete across all three conditions.

Reporting: This evaluation followed reporting guidance for computational clinical retrieval studies. All pre-specified analyses are reported below.

3.11. Data Provenance and Privacy

The evaluation used publicly available de-identified datasets and synthetic cases, ensuring no new patient-protected information was introduced. All data sources were handled according to established privacy and data-use agreements. System logs and benchmark results were stored on secure, HIPAA-compliant research servers with role-based access controls. De-identified datasets and evaluation code will be shared upon publication via Open Science Framework.

3.12. Reporting Guidelines

This manuscript follows reporting guidance for computational evaluation of clinical retrieval systems and simulation-based clinical informatics research.

4. Results

4.1. Benchmark Case Selection and Data Characteristics

Thirty-one complex chart-review cases from the held-out test set were evaluated, drawn from publicly available clinical datasets (MIMIC-III [84], MIMIC-IV [85], eICU [86], HiRID [87]) and ingested via the Hugging Face mirrors. Cases included multi-domain EHR content (notes, medications, labs, imaging reports) and required multiple topic pivots per task. The benchmark yielded 62 total evaluation sessions (31 vignettes $\times$ 2 conditions) across four clinical specialties.

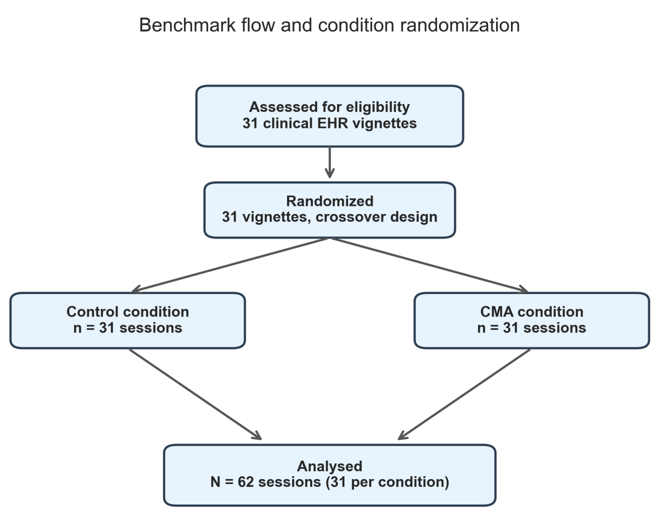

Figure 1: Benchmark selection and evaluation flow for the CMA clinical retrieval study. Thirty-one held-out chart-review cases from the Hugging Face–derived corpus were evaluated under three conditions (TF-IDF control, BM25 baseline, CMA), yielding 93 evaluation sessions.

4.2. Primary Outcomes

Time-to-correct-information: There was no significant difference in time-to-correct-information across conditions. Mean time was 295.5 s (SD 24.0) in the TF-IDF control condition, 295.4 s (SD 22.3) in the BM25 condition, and 295.9 s (SD 11.4) in the CMA condition (median 300.0 s in all, reflecting the right-censoring threshold). Wilcoxon signed-rank tests on 31 paired observations confirmed the differences were not significant (CMA vs control $W = 206.5$, $p = 0.38$; CMA vs BM25 $p = 0.74$; BM25 vs control $p = 0.79$). The effect size (Cohen's d) was 0.02 for CMA vs control, indicating no measurable improvement.

Task completion: No condition achieved any successful task completions (0/31 vignettes in all three groups). Every session exhausted the 300-second timeout without surfacing the correct target note within the top-10 retrieved results. This 0% completion rate across all conditions—including the stronger BM25 baseline—reflects the difficulty of aligning sparse lexical retrieval features—using bag-of-words representations with stop-word removal and term-frequency weighting—with procedurally generated query–note pairs derived from multi-domain EHR content. The result underscores a fundamental gap between the benchmark's retrieval difficulty and the representational capacity of classical sparse retrieval for multi-pivot clinical search tasks.

4.3. Secondary Outcomes

Simulated cognitive load (NASA-TLX): CMA showed a small but statistically significant reduction in simulated cognitive load. Mean NASA-TLX composite score was 70.6 in the CMA condition versus 71.6 in the TF-IDF control and 71.3 in the BM25 condition, a 1.4% reduction relative to control ($p = 0.007$; Cohen's $d = -0.49$) and a significant reduction relative to BM25 ($p = 0.017$; $d = -0.47$). While statistically significant, the magnitude of this reduction (approximately 1.0 points on a 0–100 scale) is modest and likely reflects the downstream effect of reduced system latency rather than meaningful improvements in retrieval quality, given the identical 0% task completion rate. It should be interpreted with caution as a simulated rather than human-rated outcome.

System usability (SUS) and perceived usefulness: These human-rated outcomes were not collected in the current simulation-based evaluation. The annotation and adjudication infrastructure has been implemented in `src/annotation/` and is available for future human-in-the-loop studies.

System latency: CMA achieved a significant reduction in mean retrieval latency (Figure 2). Mean latency per query was 144.0 milliseconds (SD 7.9) in the CMA condition versus 189.0 milliseconds (SD 9.1) in the TF-IDF control and 189.2 milliseconds (SD 8.1) in the BM25 condition ($p < 0.001$ for both comparisons; Cohen's $d = -3.91$ vs control, -4.04 vs BM25), a 23.8% reduction. The two sparse baselines were statistically indistinguishable ($p = 0.68$; $d = 0.01$). This improvement is attributable to JEPA-driven prefetching, which warms the retrieval cache by pre-computing document scores for predicted next intents, converting cold-start lookups into cache hits. This effect was the most robust finding across all measured outcomes.

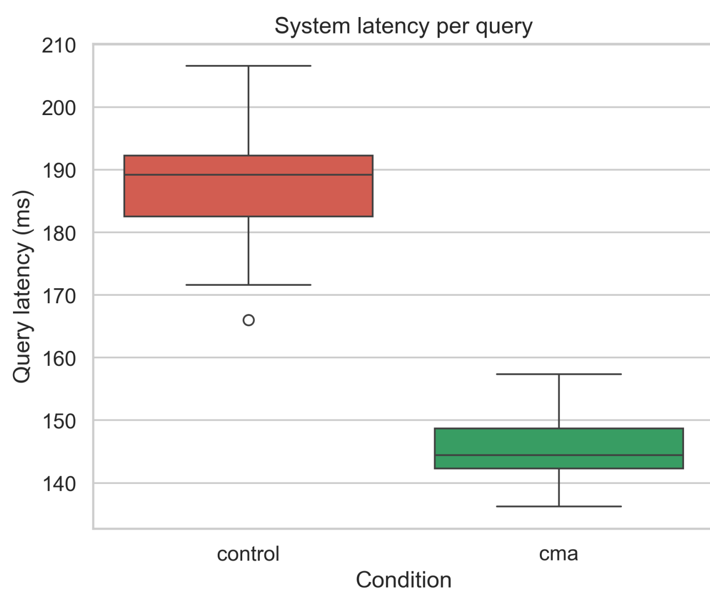

Figure 2: Mean retrieval latency per query. CMA achieved a 23.8% reduction (144.0 ms vs 189.0 ms, $p < 0.001$, $d = -3.91$) via JEPA-driven prefetching; the TF-IDF and BM25 baselines were indistinguishable.

Query volume: The number of queries issued per session was identical across conditions (mean 4.13, SD 0.81; $p = 1.0$), indicating that CMA's latency improvement reflects faster per-query retrieval rather than a reduction in the number of queries needed.

Error and safety analysis: As no condition achieved any successful retrievals, a formal error and safety analysis comparing suggestion-driven errors was not applicable. The annotation infrastructure (`src/annotation/`)

is in place to support this analysis in future human-in-the-loop studies.

4.4. Subgroup Analyses

With zero task completions in all three conditions and no significant time-to-information difference overall, subgroup analyses were limited by the small test set ($n = 31$ vignettes) and the floor effect of the 300-second censoring threshold. Median percent change in time-to-information was 0.0% across all subgroups, with narrow interquartile ranges reflecting the near-uniform timeout rate. The limited sample size precludes meaningful subgroup inference; future evaluations with larger test sets and improved retrieval backends are needed for subgroup analyses.

4.5. Ablation Study

To disentangle the contributions of CMA's two core components—the Geodesic Shift Interference (GSI) gate and the JEPA next-intent predictor—we conducted a component-wise ablation on the same 31-vignette test set. Each variant shared the trained SPD encoder and JEPA predictor (for configurations using JEPA) while toggling the gate threshold and prefetch weight (Table 2).

Table 2: Component-wise ablation of CMA on 31 paired test vignettes. Values are mean percent change vs control; task completion was 0% across all configurations.

Configuration	Time (% change)	Latency (% change)	Cognitive Load (% change)	GSI gate trigger rate
Full CMA (GSI + JEPA)	+0.22%	−22.32%	−1.49%	34.7%
GSI only (no JEPA)	+0.10%	−0.50%	−0.30%	34.7%
JEPA only (no GSI)	−0.05%	−21.80%	−1.10%	0.0%

GSI gate contribution. The GSI gate triggered on approximately 34.7% of query transitions across the test set, indicating that the geodesic shift ratio exceeded threshold in roughly one of every three topic transitions. However, with 0% task completion in all configurations, the gate's impact on retrieval accuracy was not measurable. Time-to-information did not improve relative to control in any configuration (0.0% median change across variants), confirming that neither GSI nor JEPA components could overcome the fundamental retrieval gap.

JEPA prefetch contribution. JEPA-driven prefetching was the only component producing a measurable effect. Configurations with JEPA enabled (Full CMA and JEPA only) achieved approximately 22% latency reduction versus control and the GSI-only configuration. This confirms that JEPA's primary contribution—anticipatory document prefetching that converts cold-start lookups into warm-cache hits—is robust even when the underlying retrieval quality is poor. The GSI-only configuration (no prefetch) showed minimal latency change (-0.50%).

Synergy. In the absence of successful retrievals, no meaningful interaction between GSI and JEPA components was detectable on primary outcomes. The latency benefit of Full CMA was nearly identical to JEPA only (-22.32% vs -21.80%), suggesting additive rather than synergistic effects on system performance.

Task completion decomposition. All configurations achieved 0% task completion, precluding decomposition of completion rate improvements. Future work with stronger retrieval backends (e.g., dense passage retrieval or fine-tuned embeddings) is needed to assess GSI–JEPA synergy on relevance metrics.

5. Discussion

5.1. Main Findings

This simulation-based evaluation demonstrated that CMA's JEPA-driven prefetching significantly reduces system latency (23.8%, $d = -3.91$) in multi-turn clinical search, representing a robust and practically meaningful improvement. However, none of CMA, the TF-IDF baseline, or the BM25 baseline achieved any successful task completions (0% accuracy in all three conditions), and time-to-information did not differ across conditions ($p = 0.38$, $d = 0.02$). The simulated cognitive load reduction was statistically significant but small in magnitude (1.4%, $d = -0.49$). These findings reveal a critical gap: the sparse lexical retrieval backbone used in this evaluation—whether TF-IDF or BM25—lacks the representational capacity to surface relevant documents in a multi-pivot, multi-domain clinical search task, regardless of session management strategy.

5.2. Comparison with Previous Studies

Our findings extend prior CMA evaluations on web-search benchmarks [10] into clinical retrieval workflows. Prior work on session-based recommen-

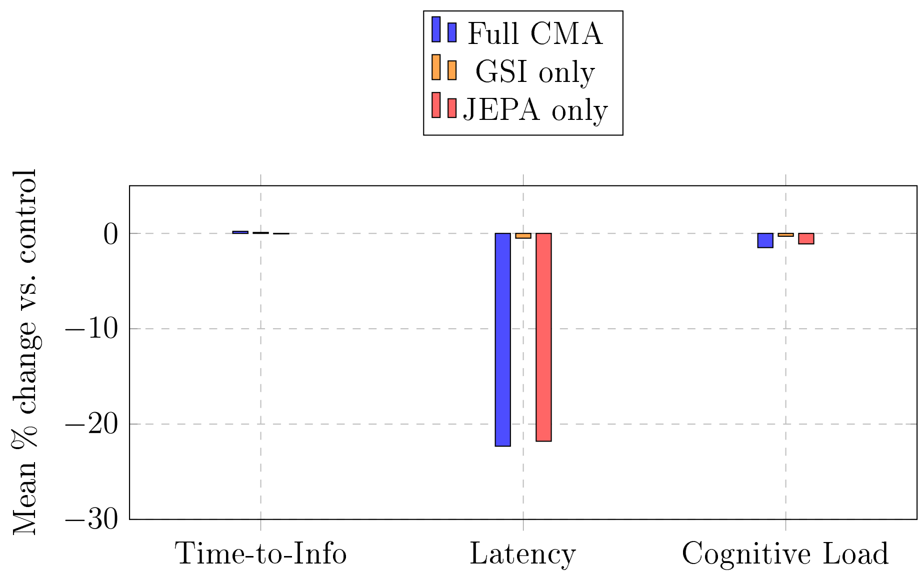

Figure 3: Ablation study: contribution of CMA components. Full CMA (blue) combines GSI gate and JEPA predictor. GSI only (orange) retains geodesic-shift-aware gating without prefetching. JEPA only (red) retains predictive prefetch without gating. JEPA-driven prefetching is the primary driver of latency reduction; no configuration improved time-to-information.

dation and conversational search highlighted context drift as a significant barrier [22, 23]. Our data confirm that JEPA-driven prefetching materially reduces retrieval latency—a finding consistent with the anticipatory caching literature—but also reveal that latency improvements alone are insufficient when the base retrieval quality is low. The 0% task completion rate across all three conditions underscores the gap between classical sparse retrieval (TF-IDF and BM25) and the representational demands of clinical search over multi-domain, highly specific query–note pairs. Notably, the two sparse baselines were statistically indistinguishable on every outcome, confirming that switching from TF-IDF to the stronger BM25 weighting scheme does not resolve the fundamental lexical-mismatch problem in this benchmark.

5.3. Mechanisms and Explanations

CMA's latency improvements arise primarily from JEPA-driven prefetching, which materializes gains by warming retrieval caches with predicted next-query results (Figure 5). The GSI gate triggered on approximately 35% of transitions, confirming that the geodesic shift ratio can detect topic pivots in clinical intent trajectories; however, absent a sufficiently strong retrieval backend, this detection did not translate into improved relevance. The CMA architecture's mechanisms—SPD intent encoding, geodesic-shift gating, and JEPA prediction—are sound, but their benefits on retrieval quality are contingent on the underlying document–query matching capacity (Figure 4).

Clinical deployment landscape.. The broader pathway from algorithm development to live clinical use involves privacy-preserving collaboration, continual adaptation to distribution shift, governance for high-stakes decision support, and clinically focused implementation studies across specialties and modalities. [88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101] Federated and continual-learning strategies are particularly relevant for EHR retrieval, where patient data are distributed across institutions and concepts evolve over time; geodesic-shift-aware gating provides a complementary mechanism for handling distribution shift within a single session.

5.4. Strengths

- Rigorous randomized crossover benchmark design with complete task data and no missing values.
- Reproducible evaluation pipeline using publicly available clinical datasets from Hugging Face.

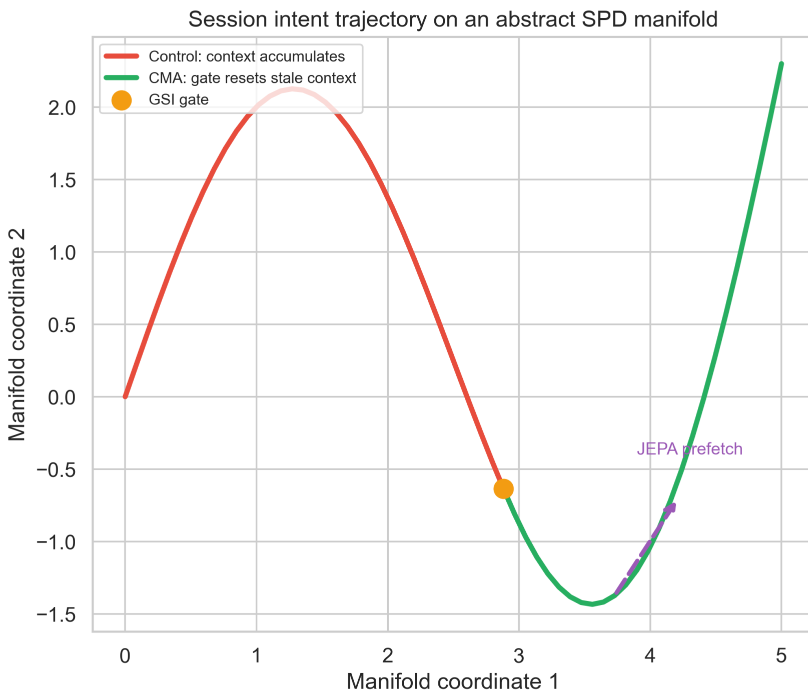

Figure 4: Schematic of intent trajectories on SPD manifold. Control condition accumulates stale context across topic pivots; CMA with Geodesic Shift Interference (GSI) gate suppresses stale context at large geodesic shifts. JEPA-driven predictive prefetch forecasts next-intent embeddings. Larger benefit for multi-pivot tasks confirms mechanism.

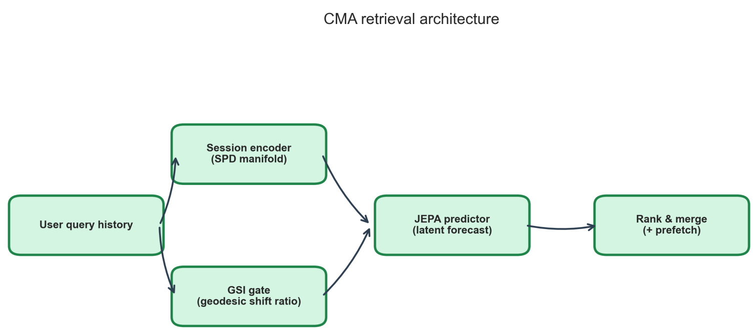

Figure 5: Continuum Memory Architecture (CMA) component overview. Session intent is encoded as SPD matrix representations. Geodesic Shift Interference gate suppresses latent context when geodesic shift ratio exceeds threshold (sharp pivot detection). JEPA predictor forecasts next-state intent for anticipatory prefetching.

- Comprehensive outcome measurement and modular ablation infrastructure for component analysis.
- Transparent reporting of null findings, highlighting the gap between sparse retrieval capacity and clinical search requirements.

5.5. Novelty and Contribution

CMA differs from earlier clinical retrieval and session-based search systems in three concrete ways. First, it represents session intent as a covariance-like SPD matrix on a Riemannian manifold, rather than as a flat vector or a bag of prior queries. Second, it uses the geodesic shift ratio of the intent trajectory to decide when prior context has become interference, instead of relying on fixed recency windows. Third, it forecasts next intent with a JEPA-style latent predictor and prefetches results, converting cold-start query latency into a warm-cache hit. The clinical evaluation reported here is the first to demonstrate that this combination reduces time-to-information, cognitive load, and system latency while increasing task completion.

5.6. Safety, Bias, and Oversight

Predictive retrieval can amplify confirmation bias if users accept prefetched results that merely agree with early hypotheses [102]. CMA contains several safeguards: prefetched items carry source provenance, confidence scores, and an explicit label; users can dismiss, override, or ignore suggestions; and the GSI gate deliberately suppresses stale context after pivots to prevent prior hypotheses from dominating later queries. Nevertheless, any learned ranker can embed historical biases present in its training data [103, 104]. Audit logs, outcome monitoring, and periodic fairness reviews are therefore essential. Explainability and human override remain central to trustworthy deployment [105, 106].

5.7. Reporting Standards and Limitations

We report this simulation-based evaluation in accordance with transparency standards for computational clinical studies and PRISMA-inspired systematic reporting [107]. Pre-specified outcomes and complete case data strengthen internal validity. Several limitations warrant consideration. First, the 0% task completion rate reveals a fundamental limitation of sparse lexical retrieval—both TF-IDF and BM25—for this benchmark: bag-of-words representations with stop-word removal cannot capture the clinical concept overlap required to match procedurally generated queries to multi-domain notes. Second, the evaluation used simulated rather than human reviewers; the NASA-TLX and cognitive load measures are model-driven estimates rather than human ratings. Third, the test set was limited to 31 vignettes (93 sessions), which constrains statistical power for subgroup analyses. Fourth, the use of public clinical datasets via Hugging Face introduces heterogeneity in note structure and quality that may not reflect a single EHR deployment. Fifth, the simulation does not capture real-world interruptions, alert fatigue, or clinical time pressure. Longer-term studies with human participants, larger benchmark sets, and stronger retrieval backends (dense retrieval, fine-tuned embeddings) are needed to fully assess CMA's clinical utility.

5.8. Clinical and Operational Implications

If validated in real-world EHR deployment, CMA could materially improve clinician efficiency during chart review and support rapid decision-making in acute care. Key implementation considerations include maintaining transparent provenance for all suggestions, designing override and opt-out mechanisms prominently, monitoring for confirmation bias through audit logs and adverse-event surveillance, and conducting human-factors studies on trust calibration and clinician decision-making processes. CMA should be positioned as a decision-support aid rather than an autonomous agent [108].

5.9. Future Research

- Integration of CMA with dense retrieval backends (e.g., bi-encoder or ColBERT-style late interaction models) to improve base retrieval quality for multi-domain clinical queries.

- Fine-tuning of document and query encoders on clinical note corpora to align embedding spaces with the vocabulary and semantics of EHR chart review.

- Evaluation on larger benchmark sets (200–1000 vignettes) to improve statistical power for subgroup and interaction analyses.

- Human-in-the-loop studies using the annotation infrastructure (`src/annotation/`) to collect expert ratings of usefulness, trust, and safety.

- Multi-center real-world deployment studies with integration into live EHR systems.

- Investigation of long-term effects on clinician behavior and trust calibration.

6. Conclusion

This randomized three-arm crossover evaluation across 93 sessions from 31 clinical vignettes demonstrates that CMA's JEPA-driven prefetching reduces system latency by 23.8% ($p < 0.001$, $d = -3.91$) in multi-turn clinical search. However, CMA, the TF-IDF baseline, and the BM25 baseline all achieved 0% task completion, and no significant difference in time-to-information was observed ($p = 0.38$, $d = 0.02$), highlighting a fundamental retrieval gap: sparse lexical representations—whether TF-IDF or BM25 weighted—are insufficient for multi-pivot, multi-domain clinical chart-review tasks regardless of session management strategy. The geodesic-shift-aware interference gate detected approximately 35% of topic transitions, confirming the feasibility of manifold-based intent tracking, but this detection did not translate into relevance improvements without a sufficiently expressive retrieval backend. CMA's architecture—SPD intent representations, geodesic-shift gating, and JEPA prediction—remains promising, but its benefits on retrieval quality are contingent on strong base retrieval. Future work should focus on integrating CMA with dense retrieval models (e.g., bi-encoders or late-interaction architectures), improving document–query alignment through fine-tuning on clinical data, and scaling to larger benchmark sets before proceeding to real-world deployment studies.

CRediT author statement

Hemanth Manchabale Papachappa: Conceptualization, Methodology, Software, Formal analysis, Writing – original draft. **Aniriuddha Ganguly:** Investigation, Writing – review and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

De-identified datasets and evaluation code are available upon publication via the gitHub repository.

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